

Figure 3: Simulation results of GreenCloud simulator

data centre data flow in real-time. It allows users to save simulation configurations as XML files and exports live results in PDF format.

The features of CloudAnalyst are listed below.

- **Graphical user interface:** Easy-to-use GUI for setting up and viewing results of all sorts of cloud computing experiments.
- **Simulation definition via a high degree of configuration and flexibility:** CloudAnalyst is equipped with modellers that have a high degree of control over the experiment by modelling entities such as data centres, virtual machines, memory, storage and bandwidth.
- **Experiment looping:** CloudAnalyst can save simulation scenarios and loop them again and again via simulation variations. It can save the results as XML files and even save PDF files of the results.
- **Efficient output:** CloudAnalyst provides graphical output of simulation results in the form of tables and charts, apart from a large amount of statistical data.

GreenCloud

GreenCloud provides a simulation environment for energy-aware cloud computing data centres. It is regarded as the most sophisticated packet-level simulator available till date for energy-aware cloud computing data centres, with a focus on cloud communications. It offers a detailed fine-grained modelling of the energy consumed by the data centre's IT equipment such as computing servers, network switches and communication links.

The GreenCloud simulator was developed by Dzmityr Kliazovich (Project Leader), research fellow at the Faculty of Science, Technology and Communication of University of Luxembourg with other team members. This simulator is used to develop novel solutions in monitoring, resource allocation, workload scheduling as well as communication protocols, optimisation and network infrastructure.

GreenCloud has been developed as an extension of the NS-2 packet-level network simulator. It distinguishes between three energy consumption components—computing energy,

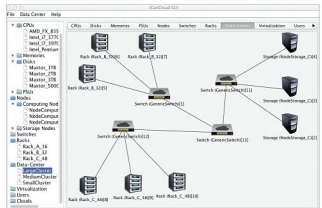


Figure 4: iCanCloud GUI

communicational energy and energy components related to the physical infrastructure of a data centre.

The latest version of GreenCloud is 2.1.2.

GreenCloud simulator calls a set of the following simulation scripts:

- **setup_params.tcl:** Contains general server configuration, switches, tasks, monitoring and migration.
- **topology.tcl:** Creates the data centre network topology.
- **dc.tcl:** Creates data centre servers and VMs.
- **user.tcl:** Defines the behaviour of cloud users.
- **record.tcl:** Sets up runtime results-reporting procedures.
- **finish.tcl:** Calculates and reports simulation statistics.

The features of GreenCloud simulator are listed below:

- The simulator mainly focuses on the cloud network and, particularly, energy consumption monitoring in cloud computing technologies.
- It supports simulation of CPU, memory, storage and networking resources.
- Supports researchers in exploring methods to minimise electricity consumption by improving power management, as well as dynamically managing and configuring the power-aware capability of the system's devices.
- Has a user-friendly GUI and is open source.

iCanCloud

iCanCloud is a cloud computing simulation platform which is based on SIMCAN and supports the simulation of large storage networks. The iCanCloud simulation framework was developed by A. Nunez and J.L. Vazquez-Poletti with the objective of predicting the trade-offs between cost and performance of a given set of applications executed in specific hardware. The simulator then provides users information about such costs. iCanCloud was designed to optimise flexibility, accuracy, performance and scalability, and has turned into a powerful simulator for designing, testing and analysing all sorts of existing and non-existing cloud architectures.

iCanCloud is being developed over the OMNeT++ platform. The latest version is 1.0 and requires OMNeT++